## Extension

Sometimes increasing the length of a code by one bit gives a new code with improved error detection or correction, which justifies the (slightly) lower rate.

**5.7.5 Definition** Let $C$ be a linear code of length $n$. If $c \in C$, let $c^*$ be the word of length $n+1$ formed by adding one extra bit to $c$ to make $\|c^*\|$ even. Let $C^* = \{c^* \mid c \in C\}$. This is a code of length $n+1$, called the *extended code of* $C$.

**5.7.8 Theorem** If a linear code $C$ has a $k \times n$ generating matrix G then the extended code $C^*$ has a $k \times (n+1)$ generating matrix

$$\mathrm{G}^* = \begin{pmatrix} \mathrm{G} & b \end{pmatrix},$$

where $b$ is a $k \times 1$ column vector such that each row of $\mathrm{G}^*$ has even weight.

**5.7.9 Theorem** If H is a parity check matrix for $C$, then a parity check matrix for $C^*$ is given by

$$\mathrm{H}^* = \begin{pmatrix} \mathrm{H} & \mathbf{1}^{\mathsf{T}} \\ \mathbf{0} & 1 \end{pmatrix},$$

where $\mathbf{1}^{\mathsf{T}}$ is a column vector containing all 1s, and $(\mathbf{0}\ 1)$ is a single extra row.

**Example** Let $C$ be the $(4,2,1)$-code given by
$C = \{0000, 1000, 0111, 1111\}$

$C^* = \{00000, 10001, 01111, 11110\}$ $(5,2,2)$-code

$$G_C = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \end{pmatrix} \rightarrow G_{C^*} = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 \end{pmatrix}$$

$$H_C = \begin{pmatrix} 0 & 0 \\ 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} \rightarrow H_{C^*} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

Let $C$ be an $(n, k, \delta)$ linear code, and let $C^*$ have parameters $(n^*, k^*, \delta^*)$. Clearly $n^* = n+1$ and $k^* = k$. Thus, the rate $r^*$ of $C^*$ is $r^* = k^*/n^* = k/(n+1)$. Notice that $C^*$ is a slightly less efficient code than $C$ since

$$r = \frac{k}{n} > \frac{k}{n+1} = r^*.$$

$$\delta^* = \|c^*\| \;=\; \begin{cases} \delta, & \text{if } \delta \text{ is even,} \\ \delta+1, & \text{if } \delta \text{ is odd.} \end{cases}$$

Thus an extended code is only useful when the distance of the original code is odd, in which case the extended code corrects no more errors than the original code, but it detects one more error.

5.7.11 Theorem Let $C$ be a linear $(n, k, \delta)$-code in which $\delta$ is odd. Then there exists a linear $(n+1, k, \delta+1)$-code.

Note that Theorems 5.7.3 and 5.7.11 give us the result that a linear $(n, k, \delta)$-code with $\delta$ odd exists if and only if a linear $(n+1, k, \delta+1)$-code exists.

Table of general binary codes: https://aeb.win.tue.nl/codes/binary-1.html
gives max # of codewords for given $n$ and even $\delta$.

### Residual

Let $C$ be a linear $(n, k, \delta)$-code and let $v \in C$ be a codeword of weight $\delta$. By considering equivalent codes, we can assume that $v$ has 1 in the first $\delta$ positions, and 0 in the remaining $n-\delta$ positions. Choose a generating matrix $G$ for code $C$, such that $G$ has $v$ as its first row. Delete the first $\delta$ columns of $G$ as well as the row of zeros that results, leaving a $(k-1) \times (n-\delta)$ matrix. The residual code $R$ of $C$ is the code with this matrix as its generating matrix.

5.7.12 Theorem Let $C$ be a linear $(n, k, \delta)$-code. The residual code is a linear $(n-\delta, k-1, \lceil \frac{\delta}{2} \rceil)$-code.

This construction was used in the proof of the Griesmer bound.

## $(a|a+b)$ Construction

Let $C_1$ be a linear $(n, k_1, \delta_1)$-code and let $C_2$ be a linear $(n, k_2, \delta_2)$-code. For each $c_1 \in C_1$ and $c_2 \in C_2$, we construct a word of length $2n$ whose first $n$ bits are the word $c_1$ and whose last $n$ bits are the word $c_1 + c_2$. These words of length $2n$ are a new linear code and we refer to this construction the $(a|a+b)$ construction. Note that $a \in C_1$ and $b \in C_2$.

Example:
Let $C$ be the linear $(4,2,1)$-code, $C = \{0000, 1000, 0111, 1111\}$
Let $D$ be the linear $(4,2,2)$-code, $D = \{0000, 1100, 0011, 1111\}$

The $(a|a+b)$ construction with $a \in C$ and $b \in D$ gives:

| | | | |
|---|---|---|---|
| 0000 0000 | 1000 1000 | 0111 0111 | 1111 1111 |
| 0000 1100 | 1000 0100 | 0111 1011 | 1111 0011 |
| 0000 0011 | 1000 1011 | 0111 0100 | 1111 1100 |
| 0000 1111 | 1000 0111 | 0111 1000 | 1111 0000 |

$n = 8$ $\quad k = 4$ $\quad \delta = 2$

The $(a|a+b)$ construction with $a \in D$ and $b \in C$ gives:

| | | | |
|---|---|---|---|
| 0000 0000 | 1100 1100 | 0011 0011 | 1111 1111 |
| 0000 1000 | 1100 0100 | 0011 1011 | 1111 0111 |
| 0000 0111 | 1100 1011 | 0011 0100 | 1111 1000 |
| 0000 1111 | 1100 0011 | 0011 1100 | 1111 0000 |

$n = 8$ $\quad k = 4$ $\quad \delta = 1$

Let $C_1$ be a linear $(n, k_1, \delta_1)$-code with generating matrix $\mathrm{G}_{C_1}$ and parity check matrix $\mathrm{H}_{C_1}$, let $C_2$ be a linear $(n, k_2, \delta_2)$-code with generating matrix $\mathrm{G}_{C_2}$ and parity check matrix $\mathrm{H}_{C_2}$, and let $C$ be the code constructed from $C_1$ and $C_2$ using the $(a|a+b)$ construction with $a \in C_1$ and $b \in C_2$. Then $\mathrm{G}_C$ and $\mathrm{H}_C$ are a generating matrix and parity check matrix, respectively, for $C$, where

$$\mathrm{G}_C = \begin{pmatrix} \mathrm{G}_{C_1} & \mathrm{G}_{C_1} \\ \mathbf{0} & \mathrm{G}_{C_2} \end{pmatrix} \qquad \mathrm{H}_C = \begin{pmatrix} \mathrm{H}_{C_1} & \mathrm{H}_{C_2} \\ \mathbf{0} & \mathrm{H}_{C_2} \end{pmatrix}.$$

5.7.14 **Theorem** Let $C_1$ be a linear $(n, k_1, \delta_1)$-code and let $C_2$ be a linear $(n, k_2, \delta_2)$-code. Then there exists a linear $(2n, k_1 + k_2, d)$-code where $d = \min\{2\delta_1, \delta_2\}$.

## Proof idea for Theorem 5.7.14

Length: 'a' has length $n$, 'a+b' has length $n$, so a|a+b has length $2n$.

Dimension: For each codeword of $C_1$, we get $|C_2|$ new codewords, so # new codewords $= 2^{k_1} \cdot 2^{k_2} = 2^{k_1+k_2}$, so dimension is $k_1 + k_2$.

Distance: Let $u$ be a word of weight $\delta_1$ in $C_1$. Then $u|u$ is a new codeword (since $\underset{\sim}{0} \in C_2$) and $\|u|u\| = 2\delta_1$ so the distance of the new code is $\leq 2\delta_1$.

If $v \neq 0$ and $v \in C_2$, then $\|a|a+v\| \geq \|v\| \geq \delta_2$

since if bit $i$ of $v = 1$ then:

| | | |
|---|---|---|
| $\underset{\sim}{v}$ | 1 | 1 |
| $\underset{\sim}{a}$ | 0 | 1 |
| $\underset{\sim}{a}+\underset{\sim}{v}$ | 1 | 0 |

Case 1 bit $i$ of $a+v$ is 0 if bit $i$ of $a$ is 1.

Case 2 bit $i$ of $a+v$ is 1 if bit $i$ of $a$ is 0.

So the new code has distance $\min\{2\delta_1, \delta_2\}$

## Exercise

Let $C_1$ be the $(7,4,3)$-Hamming code and $C_2$ be the $(7,1,7)$-code $\{0000000, 1111111\}$. What are the parameters that result from the $(a \mid a+b)$ construction if

(a) $a \in C_1$ and $b \in C_2$.

(b) $a \in C_2$ and $b \in C_1$.

(a) $(14, 5, 6)$ $\min\{2\cdot 3, 7\}$

(b) $(14, 5, 3)$

Summary of techniques:

Shortening : $(n,k,\delta) \to (n-1, k-1, \delta)$

Projection : $(n,k,\delta)$ with $\delta > 1 \to (n-1, k, \delta-1)$

Extension: $(n,k,\delta)$ with $\delta$ odd $\to (n+1, k, \delta+1)$

$\exists\,(n,k,\delta)$ with $\delta$ odd iff $\exists\,(n+1,k,\delta+1)$

Residual : $(n,k,\delta) \to \left(n-\delta,\ k-1,\ \left\lceil \frac{\delta}{2} \right\rceil\right)$

$(a \mid a+b)$ : $(n, k_1, \delta_1) + (n, k_2, \delta_2) \to (2n,\ k_1+k_2,\ \min\{2\delta_1, \delta_2\})$

## 6.1. Reed-Muller Codes

Mariner 9 was successfully launched on 30 May, 1971, and became the first artificial satellite of Mars when it arrived on 13 November, 1971 and went into orbit. It transmitted pictures back to Earth for nearly a year and completed its final transmission on 27 October, 1972.

Mariner 9 had a radio transmitter with a power of only 20 watts, and was transmitting over a distance of 84 million miles. Despite this, near-perfect pictures were obtained. Each picture transmitted by Mariner 9 is made up of over half a million tiny picture elements forming a rectangular array. Each picture element is a uniform shade of grey, the precise shade being specified by a 9-bit binary number. Thus each picture was represented using 5,250,000 bits, in grey-scale. Given the large distances, low power, poor reliability of electronics in the transmitter and receiver, and the general background noise of space, many errors occurred in transmission. Hence it was necessary to perform extensive error correction. Each message was divided into packets of 6 bits, and then each string of 6 bits was encoded into a 32 bit word. Thus a picture represented by 5 and a quarter million bits, was transmitted using over 5 times that number of bits.

The code used by the Mariner 9 was $RM(5)$, which has $64$ codewords, each of length $32$. Having distance $16$, it can correct any received word containing $7$ or fewer errors.

a first-order Reed-Muller code,
a linear $(32, 6, 16)$-code.

$\frac{7}{32} = 0.21875$, so errors in $\approx 20\%$ of any word was ok.

**6.1.3 Definition** The first-order Reed-Muller code of length $2^m$, denoted $RM(m)$, is the linear code given by the generating matrix $G_m$ whose columns are the binary representations of the integers from $2^m$ to $2^{m+1}-1$.

Binary rep.

1 | 0...001
2 | 0...010
3 | 0...011
4 | 0...100

$k \times n$

Recall that the *binary representation* of a non-negative integer $x$ is the binary sequence $x_t x_{t-1} \ldots x_2 x_1 x_0$ such that

$$x = x_t \cdot 2^t + x_{t-1} \cdot 2^{t-1} + \cdots + x_2 \cdot 2^2 + x_1 \cdot 2^1 + x_0 \cdot 2^0.$$

$RM(m) = (2^m, m+1, 2^{m-1})$

RM(1)

$$G = \begin{array}{cc} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \\ 2^1 \quad 2^2-1 \\ 2 \quad\quad 3 \end{array}$$

RM(2)

$$G = \begin{array}{c} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix} \\ 4 \quad 5 \quad 6 \quad 7 \end{array}$$

RM(3)

$$G = \begin{array}{c} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix} \\ 8 \quad 9 \quad 10 \quad 11 \quad 12 \quad 13 \quad 14 \quad 15 \end{array}$$

RM(4)

$$G_4 = \begin{array}{c} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix} \\ 16 \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad 31 \end{array}$$

Exercise: Show that RM(3) can be constructed from RM(2) (and another suitable code) using the (a|a+b) construction.

$$G = \begin{pmatrix} G_{c_1} & G_{c_1} \\ 0 & G_{c_2} \end{pmatrix}$$

**$(a|a+b)$ Construction**

Let $C_1$ be a linear $(n, k_1, \delta_1)$-code and let $C_2$ be a linear $(n, k_2, \delta_2)$-code. For each $c_1 \in C_1$ and $c_2 \in C_2$, we construct a word of length $2n$ whose first $n$ bits are the word $c_1$ and whose last $n$ bits are the word $c_1 + c_2$. These words of length $2n$ are a new linear code and we refer to this construction the $(a|a+b)$ construction. Note that $a \in C_1$ and $b \in C_2$.

Take $C_1$ to be $RM2$

$C_2$ to be all 0, all 1 code

$RM(2)$ $(4,3,2)$-code

all 0, all 1 $(4,1,4)$-code

$RM(3)$

$(8,4,4)$

$$\begin{pmatrix} 1&1&1&1&1&1&1&1 \\ 0&0&1&1&0&0&1&1 \\ 0&1&0&1&0&1&0&1 \\ 0&0&0&0&1&1&1&1 \end{pmatrix}$$

swap row $2 \to 4 \to 3$

Thm For $m \geq 2$, $RM(m)$ can be constructed from $RM(m-1)$ and the linear code of length $2^{m-1}$, dimension 1, and distance $2^{m-1}$, using the $(a|a+b)$ construction.

6.1.5 Theorem $RM(m)$ is a $(2^m, m+1, 2^{m-1})$-code.

*Proof* Since $G_m$ has $(2^{m+1}-1)-(2^m-1) = 2^m$ columns, the code $RM(m)$ has length $2^m$. The number of bits in the binary representation of $2^{m+1}-1$ is $m+1$, so $G_m$ has $m+1$ rows, which are easily seen to be linearly independent. Thus, $RM(m)$ has dimension $m+1$. Clearly, $RM(m)$ contains $0_{2^m}$ and $1_{2^m}$, and it can be shown that every other codeword in $RM(m)$ has weight $2^{m-1}$. Thus, $RM(m)$ has distance $2^{m-1}$. □

Reed-Muller codes are useful on noisy channels as their distance is large for their length. There are also fast encoding and decoding methods for the first-order Reed-Muller codes.

## 6.2. Hamming Codes ← perfect 1-error correcting $\delta = 3$

**6.2.1 Definition** For $r \geq 2$ let $\mathbf{H} = \mathbf{H}_r$ denote a $(2^r - 1) \times r$ matrix whose rows consist of all non-zero vectors of length $r$, each occurring exactly once. The corresponding linear code is called a *Hamming code* of length $2^r - 1$.

$r=2$

$$H_2 = \begin{pmatrix}1&1\\1&0\\0&1\end{pmatrix}$$

A (3,1,3)-code

$r=3$

$$H_3 = \begin{pmatrix}1&1&1\\1&1&0\\1&0&1\\0&1&1\\1&0&0\\0&1&0\\0&0&1\end{pmatrix}$$

A (7,4,3)-code

$r=4$

$$H_4 = \begin{pmatrix}1&1&1&1\\1&1&1&0\\1&1&0&1\\1&0&1&1\\0&1&1&1\\1&1&0&0\\1&0&1&0\\1&0&0&1\\0&1&1&0\\0&1&0&1\\0&0&1&1\\1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&1\end{pmatrix}$$

A (15,11,3)-code.

Thus, for all $r \geq 2$ the Hamming code with parity check matrix $\mathbf{H} = \mathbf{H}_r$ is a $(\underline{2^r - 1},\ \underline{2^r - 1 - r},\ 3)$ linear code.

Since the number of cosets is $2^{n-k} = 2^r$ and there are $2^r$ error patterns of weight 0 or 1, the SDA for a Hamming code is easy to construct.

| coset leader | syndrome |
|---|---|
| $000\cdots0$ | $000\cdots0$ |
| $\mathbf{I}_{2^r-1}$ | $\mathbf{H}$ |

Exercise: The following generating matrix for a (7,4,3)-Hamming code was used to encode message words of length 4 and the resulting codewords were transmitted over a BSC. Determine the most likely message word corresponding to the received word 1010100.

$$\begin{array}{c} (x_1x_2x_3x_4) \\ G = \end{array} \begin{pmatrix}1&1&0&1&0&0&0\\0&1&1&0&0&1&0\\0&1&0&0&1&0&1\\1&0&1&0&0&0&1\end{pmatrix}$$

Sum rows 3 and 4 1110100 is the most likely codeword

Row reduce

$$\begin{pmatrix}1&0&0&0&1&1&0\\0&1&0&0&1&0&1\\0&0&1&0&1&1&1\\0&0&0&1&0&1&1\end{pmatrix}$$

$$H = \begin{pmatrix}1&1&0\\1&0&1\\1&1&1\\0&1&1\\1&0&0\\0&1&0\\0&0&1\end{pmatrix}$$

1010100 H = 101 corresponds to row 2 of H, so mostly likely codeword is 1110100.

message word 0011

Since Hamming codes have an odd distance, it is sometimes useful to consider the extended Hamming codes. For $r \geq 2$, the extended Hamming code is a $(2^r, 2^r - 1 - r, 4)$ code.

| Code | Generating Matrix | Parity Check Matrix |
|---|---|---|

$$\begin{tikzcd}
\underset{C}{(n,k)} & \underset{G_c}{k \times n} \arrow[rr, bend left, "4.6.13"] & & \underset{H_c}{n \times (n-k)} \arrow[dll, "T"] \\
\underset{C^\perp}{(n,n-k)} & \underset{(n-k)\times n}{H_c^T = G_{c^\perp}} \arrow[rr, bend right, "4.6.13"'] & & \underset{n \times k}{G_c^T = H_{c^\perp}} \arrow[ull, "T"']
\end{tikzcd}$$

**5.7.9 Theorem** If $\mathbf{H}$ is a parity check matrix for $C$, then a parity check matrix for $C^*$ is given by

$$\mathbf{H}^* = \begin{pmatrix} \mathbf{H} & \mathbf{1}^T \\ \mathbf{0} & 1 \end{pmatrix},$$

<u>Exercise</u> Show that the dual of the extended Hamming code of length 8 is equivalent to RM(3).

ext. Ham code

$$H = \begin{matrix} 1111 \\ 1101 \\ 1011 \\ 0111 \\ 1001 \\ 0101 \\ 0011 \\ 0001 \end{matrix} \xrightarrow{\text{transpose}} G_{c^\perp} \begin{pmatrix} 11101000 \\ 11010100 \\ 10110010 \\ 11111111 \end{pmatrix}$$

swap row 1 to the top & reorder columns

Recall that the rows of the parity check matrix for a $(2^r-1, 2^r-1-r, 3)$-Hamming code are all the non-zero words of length $r$ (so the binary representations of $1, 2, \dots 2^r-1$).

To build a parity check matrix for the extended code we add a row of zeros (so now we have binary representations of the numbers $0, 1, 2, \dots 2^r-1$) and a column of 1's. Where you place the column of 1's doesn't matter, so if you place it as the first column, you have that the rows are the binary representations of the numbers $2^r, 2^r+1, \dots 2^{r+1}-1$. Take the transpose and you have a generating matrix for $RM(r)$.

**6.2.3 Definition** The dual of the Hamming code of length $2^r - 1$ is called the *simplex* code of length $2^r - 1$.

Hamming code $(2^r-1, 2^r-1-r, 3)$ $\xrightarrow[\text{add one bit}]{\text{extension}}$ Extended Hamming code $(2^r, 2^r-1-r, 4)$

Hamming code $\xrightarrow{\text{dual}}$ Simplex code $(2^r-1, r, 2^{r-1})$

pc matrix $\xrightarrow{\text{transpose}}$ gen matrix

Extended Hamming code $\xrightarrow{\text{dual}}$ First-order Reed-Muller code $(2^r, r+1, 2^{r-1})$

pc matrix $\xrightarrow{\text{transpose}}$ gen matrix.

Recall various families of binary linear codes we have seen

$(n,n,1)$ - codes all words in $\mathbb{F}_2^n$ (perfect codes)

$(n,1,n)$ - codes $\{\underset{\sim}{0}, \underset{\sim}{1}\}$ (perfect for $n$ odd)

$(2^m, m+1, 2^{m-1})$ - codes $1^{st}$ order Reed-Muller codes $RM(m)$

$(2^r-1, 2^r-1-r, 3)$ - codes Hamming codes (perfect codes)

$(2^r-1, r, 2^{r-1})$ - codes Simplex codes (dual of Hamming codes).

Use the $\underline{(a|a+b)}$ construction $\Rightarrow (2n, k_1+k_2, \min\{2\delta_1, \delta_2\})$ to show that there exists a

① $(30, 15, 6)$ - code

$C_1$ is the $(15, 11, 3)$ Hamming code

$C_2$ is the $(15, 4, 8)$ Simplex code

② $(2^{r+1}-2, r+1, 2^r-1)$ - code

$2(2^r-1)$

$C_1$ is the $(2^r-1, r, 2^{r-1})$ simplex code

$C_2$ is the $(2^r-1, 1, 2^r-1)$ all 0, all 1.

$\min\{2(2^{r-1}), 2^r-1\}$

**Question [Todd Ebert (1998)]** Suppose $n$ players enter a room and a red or blue hat is placed on each person's head. The colour of each hat is determined randomly. Each person can see all the other players' hats but not their own.

No communication is allowed. Once they have had a chance to look at the other hats, the players must each simultaneously guess the colour of their own hat, or pass.

The group wins \$1 billion if *at least* one player guesses correctly and *no* players guess incorrectly.

What is the optimal strategy?

Note that the players decide on a strategy before the game starts.

**Question [Todd Ebert (1998)]** Suppose $n$ players enter a room and a red or blue hat is placed on each person's head. The colour of each hat is determined randomly. Each person can see all the other players' hats but not their own.

No communication is allowed. Once they have had a chance to look at the other hats, the players must each simultaneously guess the colour of their own hat, or pass.

The group wins $1 billion if *at least* one player guesses correctly and *no* players guess incorrectly.

What is the optimal strategy?

Note that the players decide on a strategy before the game starts.

If one designated player guesses and everyone else passes, then the group has a 50% chance of success.

Can we do better than 50%?

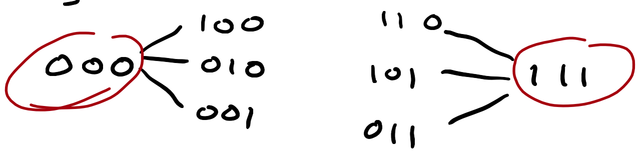
$n=3$ Strategy: Every player does the following:
If you see 2 different coloured hats – pass.
If you see 2 hats of the same colour – guess the other colour.

| Hat colour P1 | P2 | P3 | Pass/Guess P1 | P2 | P3 | Win or Lose? |
|---|---|---|---|---|---|---|
| red | red | red | blue | blue | blue | Lose |
| red | red | blue | pass | pass | blue | Win |
| red | blue | red | pass | blue | pass | Win |
| blue | red | red | blue | pass | pass | Win |
| red | blue | blue | red | pass | pass | Win |
| blue | red | blue | pass | red | pass | Win |
| blue | blue | red | pass | pass | red | Win |
| blue | blue | blue | red | red | red | Lose |

75%

What does this have to do with codes?

Consider the perfect $(3,1,3)$-code $C=\{000, 111\}$.
All words of length 3 are distance at most 1 from a codeword.




Let $0=$ red and $1=$ blue, so each hat arrangement is a binary word of length 3.

Strategy If you can make a codeword, then guess the opposite (so create a non-codeword that is distance 1 from a codeword).
If you cannot make a codeword, then pass.

Every non-codeword is a win
6 out of 8

| Hats | Guesses | Win? |
|---|---|---|
| 0 0 0 | 1 1 1 | No |
| 0 0 1 | – – 1 | Yes |
| 0 1 0 | – 1 – | Yes |
| 0 1 1 | 0 – – | Yes |
| 1 0 0 | 1 – – | Yes |
| 1 0 1 | – 0 – | Yes |
| 1 1 0 | – – 0 | Yes |
| 1 1 1 | 0 0 0 | No |

This strategy works for $n > 3$, provided you have a perfect code with $\delta = 3$.
Assign each person a bit position $1, \dots, n$.
If the hat arrangement is not a codeword, then it will be distance 1 from a codeword, so exactly one person can make a codeword and they guess (to create the non-codeword) and everyone else passes. WIN!
If the hat arrangement is a codeword, then every player guesses wrong. LOSE!

Win non-codeword $2^n - 2^k$ lose codeword $2^k$

Win prob.

$$\frac{2^n - 2^k}{2^n} = 1 - \frac{2^{2^r-1-r}}{2^{2^r-1}} = 1 - \frac{1}{2^r} = 1 - \frac{1}{n+1}$$

as $n \to \infty$
$\to 1$